Gawtam Chithra Ramesh

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Summary

A robotics researcher who takes problems from framing and approach selection through to results on physical hardware. I work across manipulation, motion planning, perception, and control in industrial R&D (ABB, Ericsson), setting the technical direction of my projects and publishing the results in peer-reviewed venues.

Education

KTH Royal Institute of Technology Aug 2024 – Jun 2026
Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0) *Stockholm, Sweden*

Indian Institute of Technology Madras Aug 2019 – Jul 2024
Dual Degree (B.Tech Engineering Design and M.Tech Robotics) (GPA 7.9/10) *Chennai, India*

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of **52** out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, ROS/ROS2, MoveIt, Gazebo, MuJoCo, PyTorch, JAX, OpenCV, Docker, URDF/SDF, CUDA

Professional Experience

Master's Thesis: Generative Motion Planning for Manipulation Jan 2026 – Jun 2026
ABB Robotics *Västerås, Sweden*

- Owned research ideation, surveying the literature and pivoting from reinforcement learning to flow matching.
- Designed a generative flow-matching motion planner guided at inference by Signal Temporal Logic robustness gradients.
- Improved spec satisfaction **31%** and success on unseen specs **75%** over SoTA planners, with no retraining.
- Transferred the approach with ABB R&D from benchmark tasks to long-horizon 6-DoF manipulation on hardware.

Device Research Intern: 3D Scene Graphs Jul 2025 – Dec 2025
Ericsson Research *Lund, Sweden*

- Integrated instance segmentation and tracking into Hydra, a real-time 3D scene-graph construction framework.
- Cut parameter sensitivity in the object layer, yielding reliable scene interpretation across diverse environments.
- Built a synchronized RGB-D inpainting pipeline that removes privacy-sensitive objects from scene graphs online.

Graduate Robotics Intern: Modular Collaborative Robots Dec 2022 – Jul 2023
Systemantics India *Bangalore, India*

- Owned the real-time system controller for a 6-DoF cobot portfolio end to end, deployed on Linux PREEMPT-RT kernel.
- Handled inter-process communication over SocketCAN and FDCAN buses using semaphores and signal interrupts.
- Implemented 7-zone S-curve trajectory generation, smoothing acceleration profiles to reduce mechanical shock.
- Integrated a ROS2 stack with MoveIt motion planning and Gazebo simulation, authoring URDF and SRDF descriptions.

Power Systems Engineer: Hyperloop Pod Battery Systems Oct 2020 – Jun 2021
Avishkar Hyperloop, IIT Madras *Chennai, India*

- Led the team that built the first custom Battery Management System in the Indian Hyperloop scene.
- Designed a high-discharge battery pack, extending pack life **10%** by cutting peak temperature 10C with heat sinks.
- Managed a 4-person team through the first hybrid setup after two withdrew, running overseas procurement.
- Won the Most Scalable Design award and placed top 24 worldwide for European Hyperloop Week in Valencia.

Research Experience

Scene Understanding for Deformable Object Manipulation Jan 2025 – Nov 2025
Prof. Florian T. Pokorny, RPL, KTH *Stockholm, Sweden*

- Coupled taxonomy-guided vision-language models with a manipulation taxonomy to map scenes onto motion primitives.
- Built a prompt and taxonomy framework turning natural-language task descriptions into robot-parsable specifications.
- Ran a quantitative failure-mode analysis across models, publishing the results at an IROS 2025 workshop.

Autonomous Ultrasound Scanning via 3D Surface Reconstruction Jan 2024 – May 2024
Nirav Patel, INSPIRE Lab, IIT Madras *Chennai, India*

- Performed 3D torso surface reconstruction from 20 predefined RGB poses via multi-view stereo on a 5-DoF robot arm.
- Segmented the abdominal ROI from the point cloud and estimated probe approach angles from surface tangent vectors.
- Closed the perception-to-planning loop with ROS/MoveIt planning, validated in Gazebo physics simulation.

Technical Projects

EKF-SLAM and State Estimation | EL2320, KTH Mar 2025 – Jun 2025

- Implemented 2D EKF-SLAM from scratch with known and unknown data association via Mahalanobis distance gating.

Robot Motion Planning and Mapping | DD2410, KTH Sep 2024 – Oct 2024

- Implemented Hybrid A* and RRT for a Dubins car; generated collision-free paths around circular and line obstacles.
- Developed a Behavior Tree mission planner for autonomous navigation, grasping, and recovery in ROS/Gazebo.

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26, ECCV26** Workshop)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25** Workshop)